

Nassim Sai

Network & Cloud Security Engineer

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Open to relocation across the EU • requires work-permit sponsorship • available immediately

PROFILE

Network and cloud security engineer, graduated 2026 with an engineering degree in Computer Systems and Network Security. My work follows one pattern: build the infrastructure, instrument it, then automate the detection layer on top — segmented enterprise networks with redundant firewalls and site-to-site VPN, NOC/SOC stacks built on Wazuh, Suricata, ELK and Zabbix, and most recently a six-service, ML-driven security observability platform for OpenShift delivered as my end-of-studies engineering project at NextStep IT. Equally at home in pfSense and Cisco configuration, Linux hardening, Python automation and Kubernetes NetworkPolicy design.

CORE SKILLS

Network Security	Cisco IOS • pfSense • FortiGate • VLAN segmentation • OpenVPN, IPsec, MPLS VPN • OSPF, BGP • firewall policy design • NAC • DNS • OpenWRT
Cloud Native & Containers	Kubernetes • OpenShift 4 • OVN-Kubernetes • NetworkPolicy • Kyverno • Docker • Podman • Fluent Bit • Microsoft Azure (in progress)
Detection & SOC	SIEM operations • Wazuh • Suricata • Snort • ELK Stack • Loki • Prometheus • Grafana • Zabbix • anomaly detection • incident response • MITRE ATT&CK • honeypots (Cowrie)
Systems & Identity	Linux (Debian, Red Hat) hardening • Windows Server • Active Directory • FreeRADIUS • VMware • GNS3
Programming & Data	Python (FastAPI, scikit-learn, Kubernetes client, NumPy) • C++ • Java • React 18 / Vite • PostgreSQL / TimescaleDB • Git
Governance	ISO/IEC 27001 ISMS • logging and traceability design • access-control policy • compliance auditing

PROFESSIONAL EXPERIENCE

Network & Security Engineer — End-of-Studies Engineering Internship Jan 2026 – May 2026
NEXT STEP IT | Chargaia 1, Tunis — hybrid

- Designed and delivered the **NetworkPolicy Intelligence System**, a six-service security observability platform for OpenShift/Kubernetes, in partnership with TEK-UP University.
- Built the full pipeline: OVN-Kubernetes ACL drop logs → Fluent Bit → Loki → flow processing → statistical and ML anomaly detection → REST/SSE API → React dashboard.
- Implemented an 8-rule NetworkPolicy audit engine (NP-001–NP-008) watching live Kubernetes objects, producing per-namespace compliance scores graded A–F.
- Ran LLM inference fully on-cluster on a dedicated tainted worker node so that **no cluster data leaves the perimeter**; measured time-to-first-token at 3.8 s.
- Published all six services as versioned container images, making the platform redeployable on any cluster without an OpenShift-internal build step.

Network Security Intern Jun 2025 – Jul 2025
Tunisie Telecom | Tunis — hybrid

- Designed and deployed a segmented enterprise architecture: 5 VLANs (DMZ, Management/NOC, Staff, HR, IT) behind a redundant high-availability pfSense firewall pair, with core/distribution/access switching layers.
- Established encrypted site-to-site OpenVPN tunnels linking headquarters to a branch office.
- Stood up the monitoring and detection layer: Suricata IDS feeding an ELK correlation stack, supervised from a Zabbix NOC dashboard covering **12+ critical hosts** with severity-classified alerting.
- Deployed Windows Server Active Directory for centralised directory and authentication services.

IoT Systems Intern Feb 2023 – Jul 2023
Compagnie des Phosphates de Gafsa (CPG) | Gafsa — on site

- Built a real-time industrial weighing and traceability system on Raspberry Pi with SIM808 cellular backhaul, streaming operational telemetry into a ThingsBoard platform for visualisation and audit.

SELECTED PROJECTS

All repositories public at github.com/nassim-sai.

NetworkPolicy Intelligence System (NPI) — flagship [nassim-sai/netpol-intelligence](#) 2026

- Six cooperating microservices: log collection (Fluent Bit/Loki), flow processing with per-namespace-pair baselines via Welford's online algorithm, three-tier anomaly detection (Z-score, deterministic rules, Isolation Forest), live policy audit, on-cluster LLM explanation service, and a FastAPI + React 18 presentation layer.
- Stack: OpenShift 4.21 (bare-metal UPI), OVN-Kubernetes, Python 3.11, FastAPI, scikit-learn, TimescaleDB hypertables, Ollama, React 18 + Vite + Recharts, Kyverno, Prometheus/Grafana.

Secure NOC & VPN Infrastructure [nassim-sai/Secure-NOC-VPN-Infrastructure](#) 2025-2026

- Enterprise telecom reference architecture shifting the defensive posture from reactive to proactive: HA firewall pair, five-VLAN segmentation, encrypted inter-site tunnels, IDS-to-SIEM correlation and a supervised NOC dashboard.

ISMS Security Architecture — NAC, Central Logging & ERP Hardening [nassim-sai/Secure-Network-Project](#) 2025

- ISO/IEC 27001-aligned design covering identity-based network access control, RADIUS/AD centralised authentication, SIEM log analysis, and the secure containerised deployment and hardening of an open-source ERP.

Network Operations Center (NOC) [nassim-sai/NOC_Project](#) 2025

- Complete open-source NOC across four VLANs, validated end to end: DHCP and VLAN connectivity tests, Zabbix performance alerting, Snort detection of port scans and ICMP floods, centralised log validation in Wazuh, and a full incident ticket lifecycle in GLPI.

Secure Portable Router [nassim-sai/Secure-Portable-Router](#) 2025

- Pocket network-security appliance: routing and firewalling on OpenWRT, DNS-level filtering, an attacker-facing honeypot for traffic inspection, and encrypted tunnelling over untrusted networks.

EDUCATION

Engineering Degree (Diplôme d'Ingénieur) — Security of Computer Systems and Networks Sep 2023 – Jul 2026

[TEK-UP University of Digital Sciences](#) | Tunis, Tunisia

- Five-year engineering cycle. End-of-studies project (PFE) completed in industry at NextStep IT.

Bachelor's Degree — Electronics, Electrical Engineering & Automation Sep 2020 – Jul 2023

[ISIMM — Higher Institute of Computer Science and Mathematics of Monastir](#) | Monastir, Tunisia

CERTIFICATIONS & TRAINING

Completed (2026)	Fortinet NSE 1, NSE 2 and NSE 3 (Certified in Cybersecurity) • Cisco Networking Academy — Introduction to Cybersecurity • Cisco Networking Academy — Networking Basics • AWS — Foundations of Prompt Engineering
In progress	Cisco Junior Cybersecurity Analyst career path (CCNA + CyberOps track) • Microsoft Learn — Azure security learning paths • Microsoft SC-900 (Security, Compliance & Identity Fundamentals)
Planned	Fortinet FortiGate Administrator (NSE 4 curriculum) • Microsoft SC-200 (Security Operations Analyst) • Red Hat RHCSA

LANGUAGES

Arabic	Native
English	Advanced — working and technical language, all documentation and interviews
French	Intermediate — professional working proficiency

COMMUNITY

Securinets TEK-UP	Active member, 2024 – present — university cybersecurity association (CTF and workshops)
IEEE ISIMM • CPU Club	Former member; graphic designer and event coordinator